

DCProject

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Benchmark Applications

Prime95

What it tests

- Prime95 mainly tests CPU stability and performance.
- It stresses the CPU heavily by calculating very large prime numbers using the Lucas-Lehmer primality test.

How it tests

- It forces the CPU to run at 100% usage for a long time.
- The program constantly performs complex floating-point calculations and integer operations, using all CPU cores.

What results it displays

- It shows whether the CPU makes calculation errors (indicating unstable hardware).
- It logs the time per iteration, temperature rise (if monitored), and pass/fail results.
- If the CPU crashes or errors out, it indicates the system is not fully stable under heavy load.

3DMark

What it tests

- 3DMark mainly tests GPU performance (graphics card), but also CPU performance in gaming-related tasks.
- It is very popular for testing how well a computer can run games and heavy graphics applications.

How it tests

- It runs intensive 3D scenes with detailed graphics (lighting, shadows, physics simulations) to stress the GPU.
- It also runs CPU physics tests to simulate things like explosions or many moving objects.

What results it displays

- A graphics score (GPU performance).
- A CPU score (CPU physics calculation speed).
- A combined score that shows overall gaming capability.
- Frame rates (FPS) during each test.
- It allows comparison of your PC to others online.